

- 7 (a) hadron not a fundamental particle/lepton is fundamental particle
or
 hadron made of quarks/lepton not made of quarks
or
 strong force/interaction acts on hadrons/does not act on leptons B1 [1]
- (b) (i) proton: up, up, down / uud B1
 neutron: up, down, down / udd B1 [2]
- (ii) composition: $2(\text{uud}) + 2(\text{udd})$
 $= 6 \text{ up, } 6 \text{ down} / 6\text{u, } 6\text{d}$ B1 [1]
- (c) (i) most of the atom is empty space
or
 the nucleus (volume) is (very) small compared to the atom B1 [1]
- (ii) nucleus is (positively) charged B1
- the mass is concentrated in (very small) nucleus/small region/small volume/small core
or
 the majority of mass in (very small) nucleus/small region/small volume/small core B1 [2]